How to use: smFISH_statistics_superplots.py

Control vs protease and DIV9 vs DIV27 for every probe and compartment, with two statistical analyses side by side, SuperPlots and an Excel master file. pipeline version 4 (October 2026).

Input: `_density_results/counts_per_image.csv` from Fish_spots_batch_v4.py. **Also used by:** the compartments, dendrite-vs-axon, advanced-quantification scripts and the plot editor (keep it in the same folder).

1. Running it

All scripts need Python 3 with: `pip install numpy pandas scipy scikit-image tifffile imagecodecs matplotlib openpyxl statsmodels`. `imagecodecs` is required to read the LZW-compressed Ilastik maps; without it every image fails in pass 1. Keep all scripts in ONE folder - several import each other.

```
python smFISH_statistics_superplots.py
-> select _density_results\counts_per_image.csv
```

Results go to `_density_results/statistics_and_plots/`. The screen summary lists the number of significant comparisons per analysis and any flagged area changes.

2. The two analyses (both always run)

	t-tests (folder t_tests)	Robust linear model (folder robust_linear_model)
Unit	replicate mean of the images	replicate mean of the images
Control vs protease	paired t-test on log10 means, per probe	one model per compartment × timepoint across all probes: mean ~ probe × condition + differentiation; contrast per probe
DIV9 vs DIV27	paired where all replicates share differentiations, otherwise Welch; tested in TIMEPOINT_CONDITIONS	same model with timepoint; redone DIV9 samples coded as own differentiation
Reported	ratio, 95% CI, p	ratio, 95% CI, p

Which to report is your decision; `COMPARISON_ttest_vs_robust_linear_model.xlsx` shows both for every comparison. Report the method you use in the methods section; the "methods" sheet in each Excel file gives the exact wording of what was computed.

3. Settings

Setting	Default	Meaning
PROBE_ORDER, COMPARTMENT_ORDER, …		order in tables and plots
REPLICATE_SUMMARY	'mean'	images → replicate value ('mean' or 'median')
NOT_SAME_DIFFERENTIATION	5 entries	(probe, replicate) whose DIV9 sample was redone from another differentiation → unpaired DIV test. **Check against your notes.**
TIMEPOINT_CONDITIONS	['protease']	conditions in which DIV9 vs DIV27 is tested
RLM_TRANSFORM / RLM_NORM / RLM_DF	'none' / HuberT / 'residual'	model details; RLM_DF='asymptotic' if your reference R analysis reports df = Inf
MULTIPLE_TESTING_CORRECTION	False	True = Benjamini-Hochberg within each comparison type
DENSITY_FACTOR / DENSITY_LABEL	100	unit only (per 100 µm²); statistics unaffected
AREA_FLAG_LIMITS	(0.8, 1.25)	area check: protease/control area ratio outside → CHECK
STYLE		all plot settings (colours, sizes, significance display, y limits per panel, …) - easier with the plot editor

4. Outputs

File (statistics_and_plots/…)	Content
t_tests/ and robust_linear_model/	per analysis: three stats CSVs, smFISH_master.xlsx, SuperPlots (PDF/PNG), panels/ (one file per panel)
stats_control_vs_protease.csv	ratio protease/control, CI, p per probe × timepoint × compartment
stats_DIV9_vs_DIV27.csv	ratio DIV27/DIV9, CI, p; test column says paired or Welch
stats_area_control_vs_protease.csv	**quality check**: compartment areas should not change with protease
smFISH_master.xlsx	per image, replicate means, all statistics, Origin-style wide sheets, methods
COMPARISON_ttest_vs_robust_linear_model.xlsx	both analyses in one row per comparison

5. Reading the results

- n = differentiation (3), not images. Never test on images as n (pseudoreplication).
- With n = 3, report the ratio and its 95% CI next to p - a consistent 1.5× effect can be non-significant.
- Area CHECK flags: the density change for that probe/compartment may partly come from the area.
- Uncorrected p-values (default) must be stated as such in the methods.

Troubleshooting

Message / symptom	Cause and fix
"missing columns ... is this counts_per_image.csv from v4?"	select the v4 counts file
"Please install statsmodels"	`pip install statsmodels`
A comparison has no p-value	fewer than 2 replicates with data (e.g. zero density) - see n column
Plots too compressed	use the plot editor (y range per panel, percentile zoom)